KANNAPPAN CHIDAMBARAM

Professional Summary:

- Backend Polyglot Engineer with 11 years of experience specializing in designing and delivering low-latency, high-performance applications within distributed architectures. Proven expertise in building robust backend services, SDKs, and microservices architectures following best practices in object-oriented design and cloud-native development.
- Demonstrated expertise in leveraging cutting-edge tools including Elasticsearch, Redis (with Redis Pub/Sub), Kafka, RabbitMQ, and CitusDB to enable real-time data processing, messaging, and fault-tolerant systems. Proficient in container orchestration with Kubernetes and cloud-native messaging services like AWS SQS, driving automation and seamless integration in cloud environments.
- Adept technical lead, experienced in steering Agile teams through complex problem-solving, mentoring developers, and promoting best practices in object-oriented design and distributed system architecture. Strong background in building robust CI/CD pipelines, automation frameworks, and secure infrastructure using Docker and Teleport, with a growing focus on DevOps principles.
- Expertise in diagnosing and resolving complex performance bottlenecks to ensure scalable, resilient, and fault-tolerant backend systems. Skilled in building robust and resilient microservices using Spring Boot, enabling fault isolation and seamless scalability.

Technical Skills

Languages:	Java 17, Python, Golang, JavaScript (ES6+), OCaml
Cloud & Orchestration:	Kubernetes, AWS SQS, Teleport
Messaging & Streaming:	Kafka, RabbitMQ, Redis Pub/Sub
Databases:	CitusDB, Cassandra, PostgreSQL, Redis Queue
Search Platform:	Elasticsearch
Frameworks:	Spring Boot, Microservices architecture, Event Sourcing, CQRS
Containerization & CI/CD:	Docker, Jenkins, GitHub Actions
Workflow Orchestration:	Temporal
Frontend:	ReactJS

Contact

Mail Id : kannappanchidambaram@gmail.com

Mobile: +91-9500853473

Contributions

- As a blogger, I have written blogs ranging from various tech interests. (https://kannappanchidambaram.medium.com/)
- As an Open source enthusiast, I have actively contributed to fixing issues in open source libraries.

Freshworks Experience:

Lead Software Engineer – *freshworks* **Jun 2024 – Present**

- Spearheaded the development and optimization of low-latency modules in the Freshchat application, accelerating message delivery speeds by 30%, significantly enhancing real-time user interactions via websockets, and drastically reducing customer wait times in high-traffic environments.
- Diagnosed and resolved complex inbox performance bottlenecks through end-to-end analysis, decreasing system latency by 40%, which resulted in improved throughput, faster data processing, and measurable uplift in user engagement and retention rates.
- Led the design and implementation of key Freshchat features such as message threading and quick replies, driving a 10% growth in daily active users and elevating customer satisfaction through streamlined communication workflows and enhanced UX.
- Designed and deployed a robust distributed lock system to enforce strict FIFO ordering for high-frequency entity updates, eliminating data conflicts and reducing concurrency-related failures by 90%, thereby ensuring system stability and reliability during peak traffic spikes.
- Implemented key features in the reports module using Python and microservices architecture, applying object-oriented programming principles to enhance module functionality and reliability; contributed to automating report generation and streamlining data aggregation, which reduced manual reporting efforts by 50% and accelerated insights delivery

Ideas2IT Experience:

Platform R&D Engineer – *Element5* **Jun 2022 – June 2024**

- Responsible for designing and developing automation platform in python from idea inception to production deployment. Collaborated in brainstorming, design, development, testing, deployment, production support and future platform roadmaps
- As part of R&D activity explored different open source solutions and extensively studied its capabilities, performance, license, support and library ecosystem, and documented this study as report and presented it to the senior architect and CTO

Tech Stack:
Python, Java, Temporal, Robotframework, Kafka, AWS (CFT, EKS, ECS, Batch, Dynamodb, S3, Parameter Store)

Platform SDK Engineer – *Element5* **Jun 2022 – June 2024**

- Developed SDK Libraries in multiple languages like java and Python for custom task sdk protocol
- Implemented 12-factor app principles while building the automation platform
- Collaborated with team in designing and developing the custom workflow orchestration sdk library using temporal and java
- Developed Custom bot automation platform using python, robotframework and playwright
- Built custom scaffolding project in python and java to bootstrap the automation of new workflows
- Built automation bot linter to support custom rules for following organization styles and code guidelines. Had the opportunity to learn more about visitor design pattern and its real-world usage.
- Developed distributed tracing solution using datadog in bot automation platform
- Introduced best coding practices in project such as code linter check, pre-commit check, code coverage, code styles, readme driven development
- Developed range of ecosystem/ suite tools needed for workflows developed with platform - Scaffolding , Developer kit , Linter
- Developed custom screen recording feature using python and opencv to capture bot automations.
- Built scripts for provisioning new developer environment setup (windows/ubuntu)

Tech Stack:
Java, Python, Temporal, Kafka, Dynamodb, S3, Gradle, Jfrog, Javascript, Datadog, AWS

Platform Devops Engineer – *Element5* **Jun 2022 – June 2024**

- Built custom CI pipeline using github actions which executes gradle build, Jfrog artifactory publish, ECR image push, acceptance testing, linter checker, yaml schema validations
- Designed and developed CD pipelines using EKS to support deployment of multiple workflows and tenant combination
- Developed a custom dockerization solution for running automation in docker containers
- Built custom configs generation python script to deploy the configs as environment variable through helm Chart

Tech Stack:
Docker, Kubernetes, AWS Services (CFT, EKS, ECS, Batch, Dynamodb, S3, Parameter Store), Helm Chart, Jenkins, Github Actions, Python, Shell scripting

Technical Lead – *Netsmart* **Jun 2021 – Jun 2022**

- Had the opportunity to learn OO best practices and design backend systems for Netsmart EHR Portal Services.
- Developed critical features for an Electronic Health Records (EHR) portal, utilizing advanced Java techniques.
- Utilized Java 8 functionalities to implement robust and efficient code for seamless integration within the EHR portal.
- Deployed Flyway for database schema versioning and migration, ensuring smooth and automated updates across different environments.

Tech Stack:
Java 8, AWS, Flyway, Docker, Kubernetes, EKS, Hibernate

Wipro Experience:

Senior Software Engineer – Hewlett Packard*(Chennai)* **Dec 2020 – June 2021**

- Had the opportunity to work on ReactJs by building a frontend application.
- Had played role as technical lead on deal management system for Hewlett packard
- Worked collaboratively in a cross functional team with product owner to design and develop backend and frontend application for Deal Management Application. (ReactJs, Spring Boot API)

Tech Stack:
AWS S3, EKS, Dynamodb
Java, ReactJs, Javascript, Postgres, Spring Boot, JPA

Senior Software Engineer – *Wipro(Tokyo)* **Aug 2017 – Dec 2020**

- Had the opportunity to learn and apply design patterns, collections and concurrency
- Had the opportunity to work extensively in multiple streams, as Java Full stack Developer and as Onsite Integration lead
- Onsite Integration Lead at SAP Finance System Integration Project
- Led onsite integration efforts for SAP Finance System, custom printing solutions, and custom vehicle management application.
- Integrated 18 different systems with DMS portal, achieving seamless interoperability.
- Successfully rolled out the project in Japan, ensuring smooth deployment.
- Designed and developed backend REST-based web services within the Spring Boot.
- Enhanced system functionality and interoperability, resulting in a 30% increase in data exchange efficiency.
- Developed an application framework for the Parts Query Management Application using Spring Boot.
- Optimized query processes, reducing response time by 25%.
- Realtime API and Batch Job Developer for Dealer Management System
- Played a pivotal role in integrating real-time APIs and batch jobs for the Dealer Management System (DMS).
- Built real-time APIs to integrate with SAP FICO, enabling instant data synchronization.
- Developed JMS Queue-based integrations in Java, ensuring efficient message handling.
- Engineered batch jobs to process chassis ordering and manage parts inventory efficiently.
- Utilized SOAP-based services in Java to facilitate seamless communication between systems.
- Implemented Java-based web services, enabling effective data exchange
- Wrote React.js scripts to create custom elements in the user interface.
- Improved UI flexibility and user experience, resulting in a 20% decrease in user error rates.

Tech Stack:
AWS S3, ECS, Dynamodb, Lambda, SQS, SNS, AWS API Gateway, CloudFront, Route53
Java, ReactJs, Javascript, Postgres, Spring Boot, JPA

Senior Software Engineer – *Mastercard (Chennai)* **Apr 2014 – Aug 2017**

- Developed migration script to process over 10 million contacts/address records from Salesforce CRM to Siebel CRM system utilizing PL/SQL procedures, resulting in a streamlined data transfer process.
- Developed Java REST APIs to seamlessly integrate with various third-party systems, facilitating the retrieval of contact, lead-gen, and opportunity details, leading to a 30% reduction in data retrieval time.
- Designed and developed python scripts to automate manual processes, increasing operational efficiency by 40%. Reduced time efforts in data validation, file manipulation, and report generation

Tech Stack:
AWS S3, EKS, Dynamodb
Java, Python, PL/SQL, Oracle 11g, Spring Boot

Education

Velammal College Of Engineering & Technology, India 2009 – 2013

- B.E Electrical & Electronics Engineering